SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTD215

Examination - July Semester 2023

Application of C++ in Multimedia

Tuesday, 21 November 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **EIGHT (8)** printed pages (including cover page).
2. You must answer **FOUR (4)** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Implement a C++ program and explain the syntax and use of data structures such as structures and class. These data structures form part of the data members and constructors in a C++ class.

Question 1a

Define a structure named Car using the C++ language to hold

- A string representing the brand of the car.
- A string representing the model number of the car.
- A float array representing the length and width of the car.
- An integer representing the seat number of the car.
- A bool representing whether its Open Market Value (OMV) exceeds S$40,000.

Do note to add comments of each data member to explain to users.

(10 marks)

Question 1b

Implement a main function to demonstrate the use of the structure defined in question 1a. The function declares an array of seven cars. Define the second element in the array as follows:

brand : "Benz",
model_numer: "CLA180"
dimensions : {4.688, 1.83},
seat_number : 5,
isOMVExceeds40k: true

(8 marks)

Question 1c

If we convert the structure created above into a C++ class, write the following based on the above `Car` class and explain coding blocks with the syntax and structure of C++ language with comments:

i. A default constructor
(You should use valid default values for some data members)

ii. A destructor

(7 marks)

Question 2

Identify suitable coding modules below.

```
1   #include <iostream>
2   using namespace std;
3
4   //To comment
5   int main() {
6       //To comment
7       int rents[4][2] = {
8       { 400, 450, 510 },
9       { 500, 560, 630 }
10      { 625, 676, 740 },
11      { 1000, 1250, 1600 } };
12
13      int floor, bedrooms;
14      cout << "Welcome to rental price checking system"  << endl;
15
16      //To comment
17      cout << "What floor do you want ? (Available floor: L1~4)"  << endl
18      cin >> floor;
19
20      //To comment
21      cout << "How many bedrooms you want ? (Available bedrooms: 1~3)"  << endl;
22      cin >> bedroom;
23
24      //To comment
25      cout << "The cost is S$" << rents[floor][bedrooms] << endl;
26      return 0;
27  }
```

Question 2a

Explain what the function of the program is and write down the answer as a comment for the comment line (Line 4).
Explain the function of other code blocks and write down the answers as comments in Line 6, 16, 20, and 24.

(10 marks)

Question 2b

There are **FIVE (5)** errors/bugs hidden in the code. Identify them and write down their respective solutions.

Error/Bug	Solution
Line number ? what error?	
Line number ? what error?	
Line number ? what error?	
Line number ? what error?	
Line number ? what error?	

(10 marks)

Question 2c

Imagine you are the user and put down one reasonable input for line 17 and line 21, write down the corresponding output of the program in a correct order.

(5 marks)

Question 3

Apply the C++ language into Monopoly Game OL. You are asked to implement a C++ file named player.cpp that has a player class.

Question 3a

Create a C++ class called Player that contains as follows:

i. An integer representing the ID of the player
ii. A string representing the name of the player
iii. A float number representing the cash the player has in real-time with initial value of 10,000.00
iv. A bool indicating if the player is bankrupt or not
v. A dynamic location large enough to store all the assets the player is holding. The location is reference by a pointer `string* assets`

(7 marks)

Question 3b

Create a Constructor of this class with initialization of aforementioned items.

(8 marks)

Question 3c

Create a C++ function to carry out the following methods in the C++ class.

- GameOver_Checker(payment) -> Check if the player is bankrupt by checking sum-up of cash and asset value is less than the payment. Display text "Game Over" if it is true. Do remember to comment to explain and elaborate the steps before delivering codes.

(10 marks)

Question 4

Appraise the use of SDL for geometry rendering. SDL has primitive functions for drawing point, lines and rectangles. Write **THREE (3)** methods for question 4a, 4b and 4c.

```
#define SDL_MAIN_HANDLED
#include <SDL.h>

const int SCREEN_WIDTH = 640;
const int SCREEN_HEIGHT = 640;

SDL_Window* gWindow = NULL;
SDL_Renderer* gRenderer = NULL;
```

```
void drawTrapezoid() {
  //Your method goes here
}
void drawStar() {
  //Your method goes here
}
void drawCircle() {
  //Your method goes here
}


int main(int argc, char* args[]) {
    SDL_Init(SDL_INIT_VIDEO);
    SDL_SetHint(SDL_HINT_RENDER_SCALE_QUALITY, "1");
    gWindow = SDL_CreateWindow("Test", SDL_WINDOWPOS_UNDEFINED,
        SDL_WINDOWPOS_UNDEFINED, SCREEN_WIDTH, SCREEN_HEIGHT,
        SDL_WINDOW_SHOWN);
    gRenderer = SDL_CreateRenderer(gWindow, -1,
SDL_RENDERER_ACCELERATED);

    bool quit = false;
    SDL_Event e;

    while (!quit) {
        while (SDL_PollEvent(&e) != 0) {
            if (e.type == SDL_QUIT)
                quit = true;
        }

        SDL_SetRenderDrawColor(gRenderer, 0xFF, 0xFF, 0xFF, 0xFF);
        SDL_RenderClear(gRenderer);
        SDL_SetRenderDrawColor(gRenderer, 0x00, 0x00, 0x00, 0xFF);
      // drawStar();
      // drawTrapezoid();
      // drawCircle ();


        SDL_RenderPresent(gRenderer);

    }

    SDL_DestroyRenderer(gRenderer);
    SDL_DestroyWindow(gWindow);
    gWindow = NULL;
    gRenderer = NULL;
    SDL_Quit();

    return 0;
}
```

Write the implementation for drawTrapezoid, drawStar and drawCircle function to produce each of the following output respectively:

Question 4a

Write the implementation for drawTrapezoid to draw a trapezoid (irregular rectangle). Alternatively, you can write down the pseudo-code to elaborate your idea of implementations by steps.

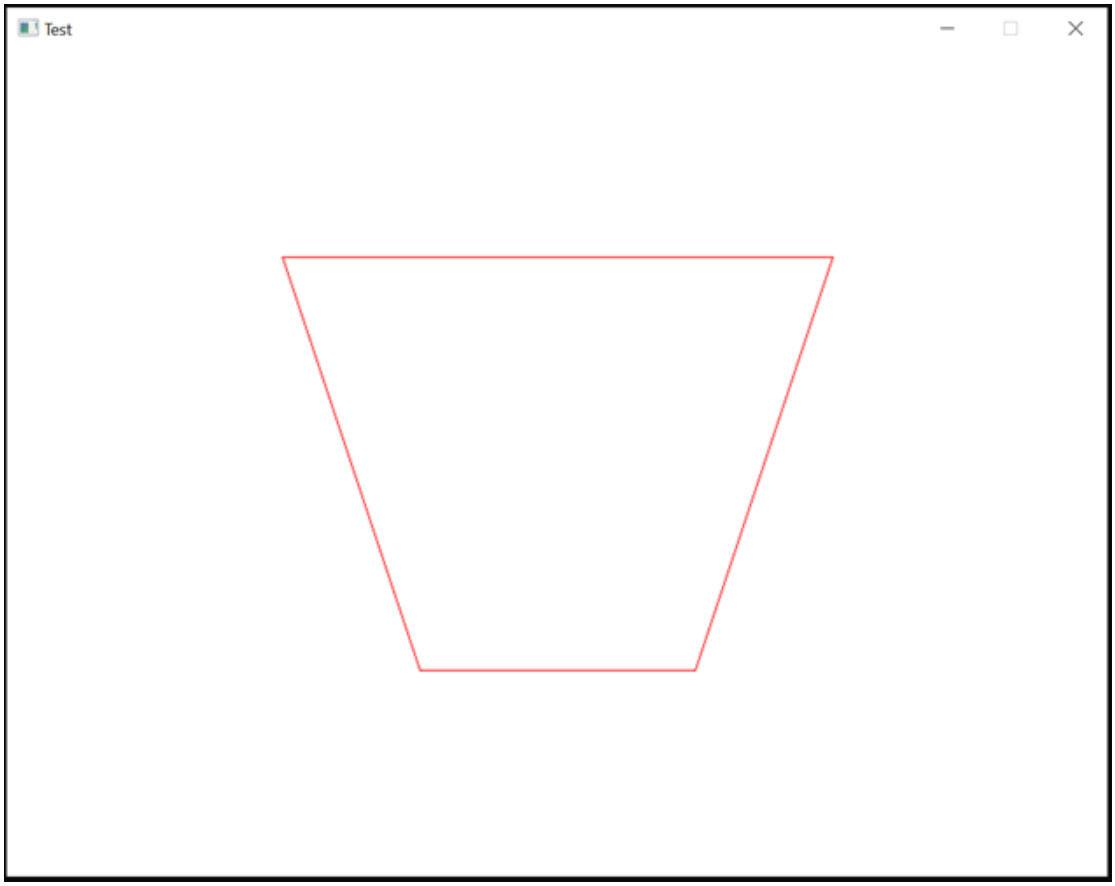

Figure 4a

(10 marks)

Question 4b

Write the implementation for drawStar to draw a star shape.
Alternatively, you can write down the pseudo-code to elaborate your idea of implementations by steps.

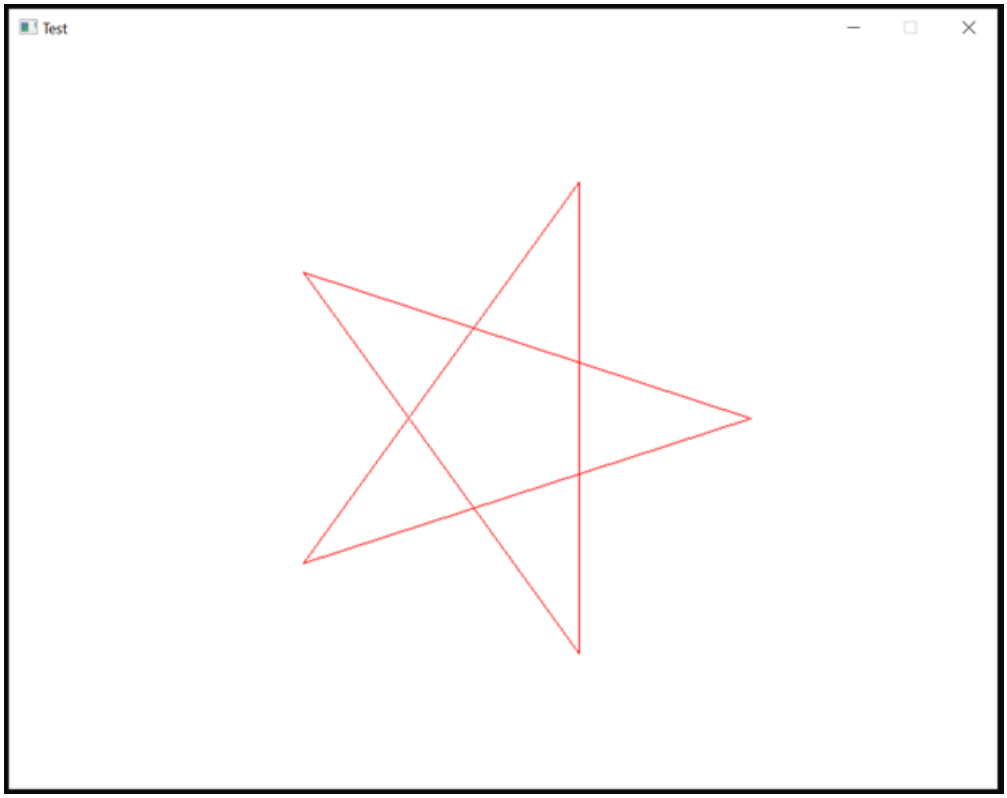

Figure 4b

(10 marks)

Question 4c

Write the implementation for drawCircle to draw a circle.
Alternatively, you can write down the pseudo-code to elaborate your idea of implementations by steps.

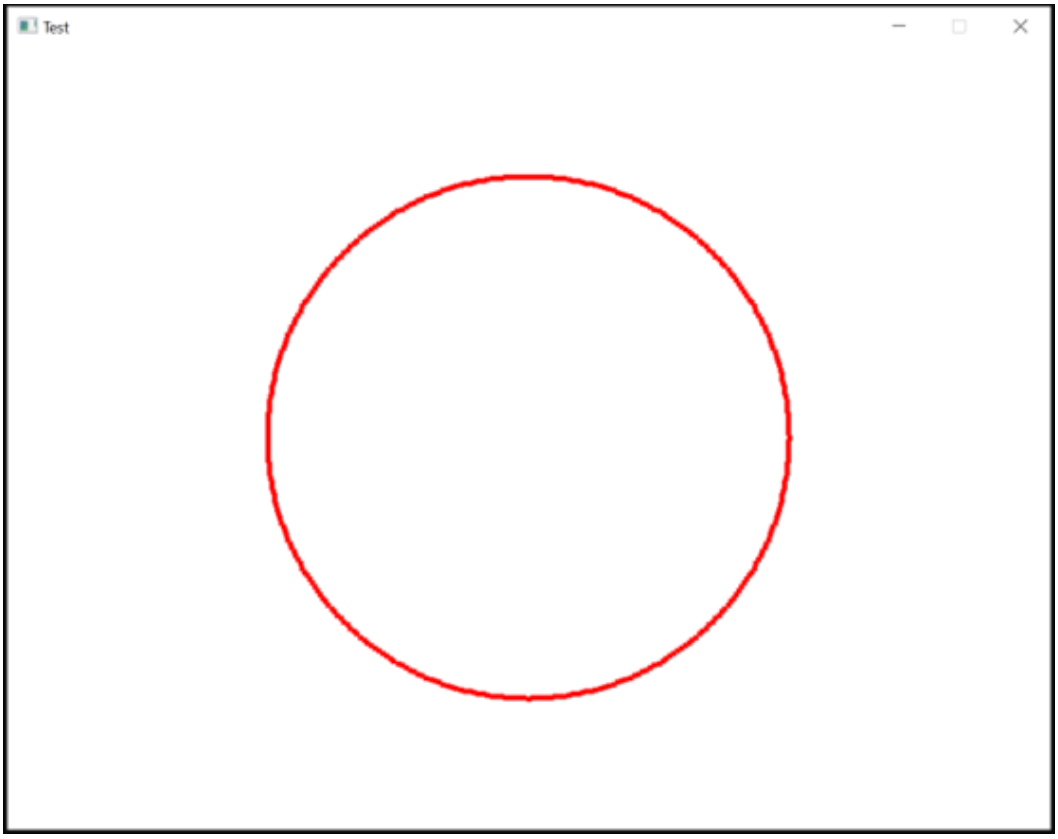

Figure 4c

(5 marks)

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